

The Curse of Dimensionality

Factorized Transitions

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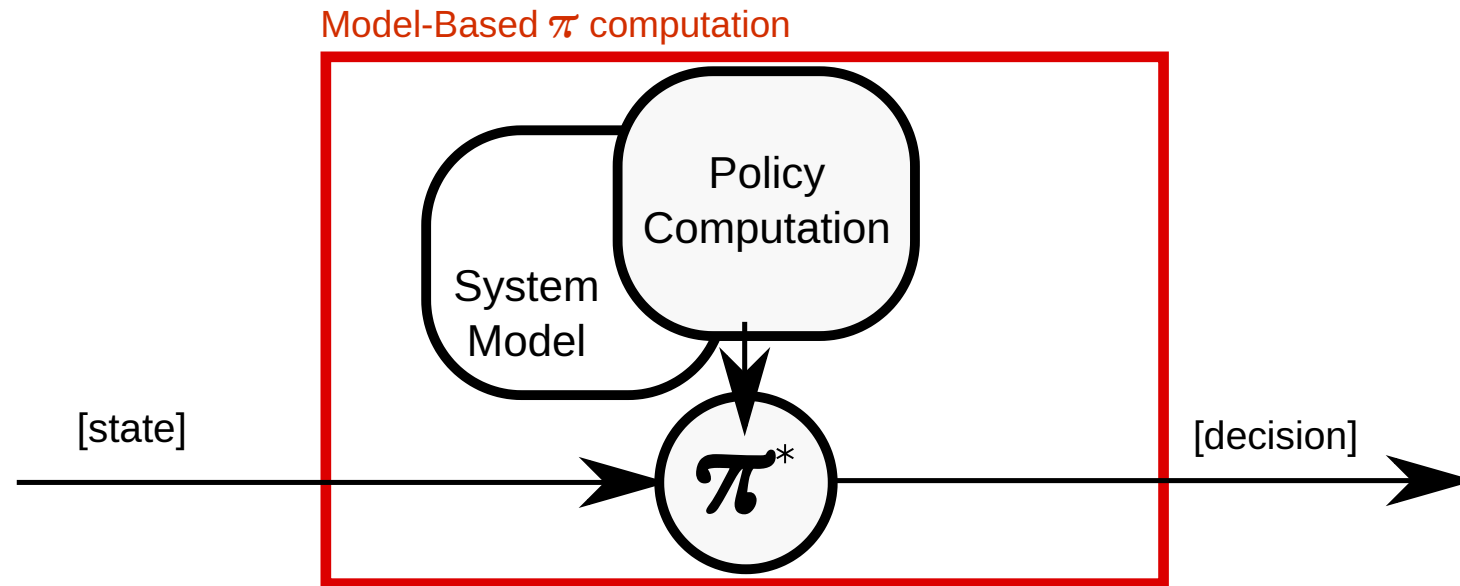
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Before to go...

Optimal decision-making requires a model.



Basic Dynamic Model:

- ▶ a **Transition** function : $T(s, a, s') \rightarrow [0, 1]$

A potentially very huge object...

1. **Bayesian Network**
2. **Factorized Transition Function**

Bayesian Network: General idea

Variables are not necessarily correlated over all the others.

Example:

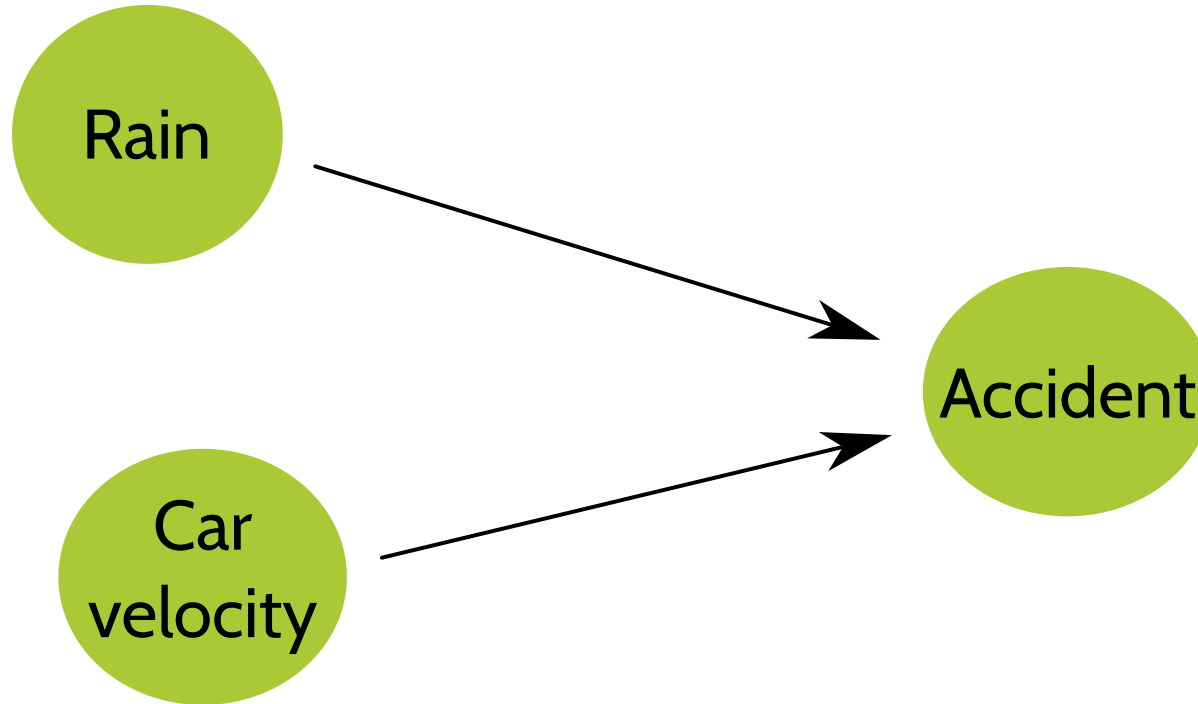
- ▶ The probability of *Rain* does not depend from *Car Velocity*.
- ▶ The *Car Velocity* does not depend from *Rain*.
- ▶ However, the probability of *Accident* depends from both *Rain* and *Car Velocity*.

Bayesian Network:

A probabilistic graphical model that represents a set of variables and their conditional dependencies

Bayesian Network: Car accident Example

Graphical model: *Node*: variable, *Orriented Edge*, dependancy



ATTENTION: *Correlation* is not *Conditional Dependancy*

Bayesian Network: Node's Probability Table

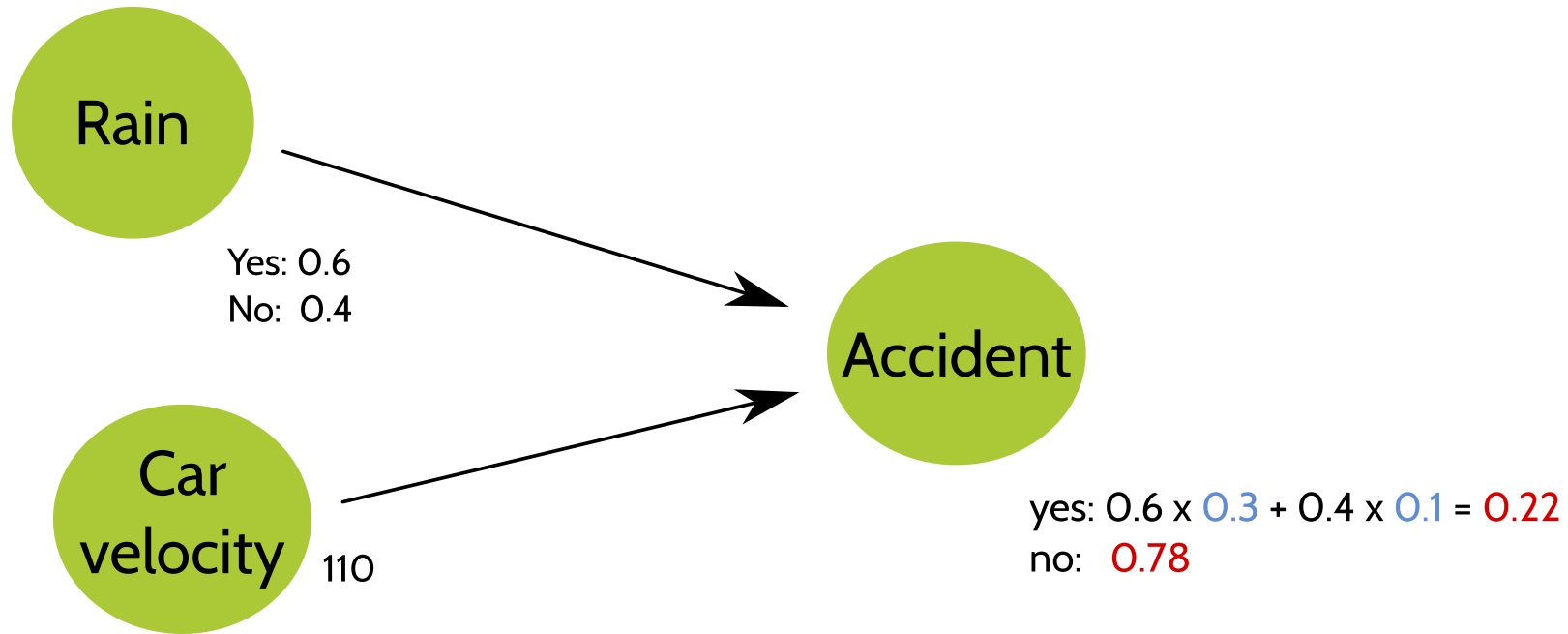
Propbality dependencies are defined node by node.



Rain	C.Vel.	Yes	No
Yes	50	0.1	0.9
	70	0.2	0.8
	90	0.2	0.8
	110	0.3	0.7
	130	0.8	0.2
No	50	0.05	0.95
	70	0.1	0.9
	90	0.1	0.9
	110	0.1	0.9
	130	0.3	0.7

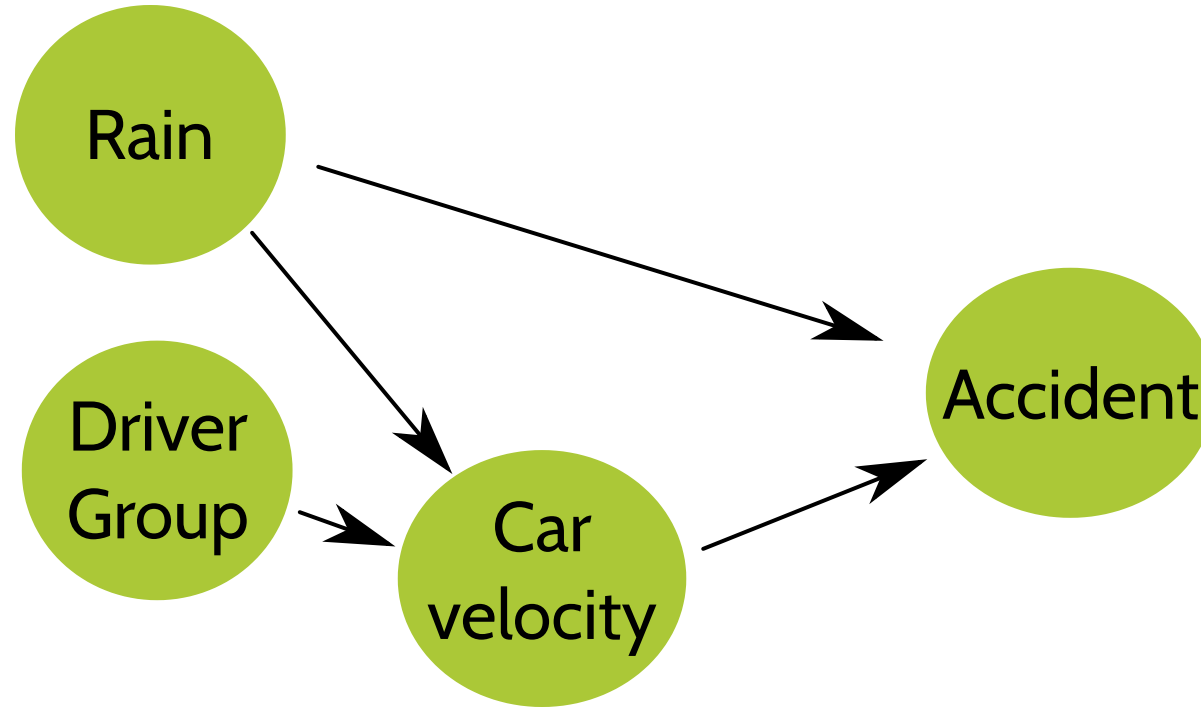
Bayesian Network: Inference

Resulting probability is computed recursively



Bayesian Network: Level-up

Growing the graph → no consequences over existing nodes



ATTENTION: *Correlation* is not *Conditional Dependancy* → **Directed Acyclic Graph**

Bayesian Network: Toolbox

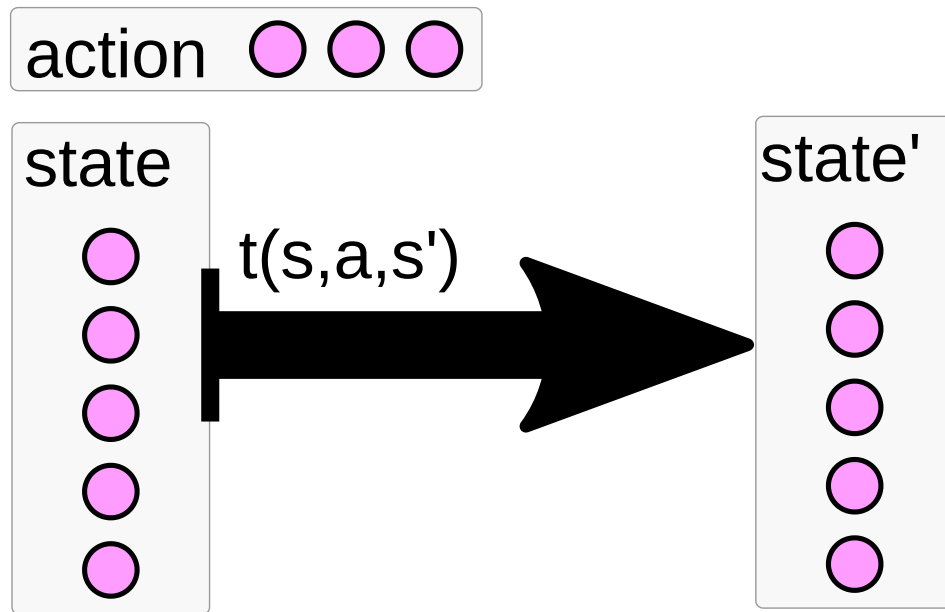
- ▶ Bayesian Network [on Wikipedia](#)
- ▶ In python - numerous implementations
 - [pomegranate](#) - define compute inferences and more.
 - [bnlearn](#) - Learning the bayesian structure (ie. detect the dependencies)
- ▶ Professionals: [bayesia](#)

1. Bayesian Network

2. **Factorized Transition Function**

Factorized Transition Function: Problem

Classically a multi-variables probabilistic evolution problem

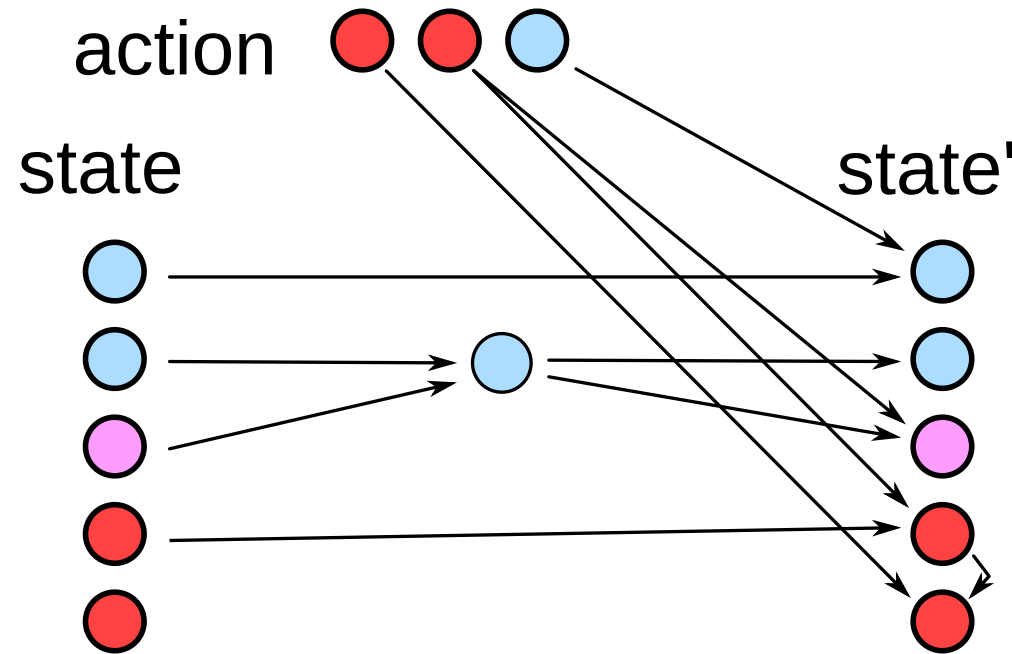


$$T(s, a, s') \rightarrow [0, 1]$$

So, it is about

$\text{Domain}(Var)^{|Var_S| \times 2 \times |Var_A|}$
values

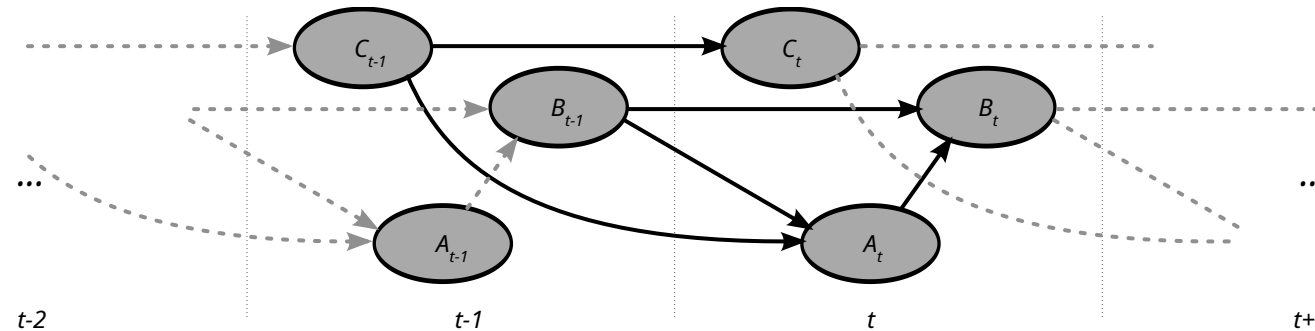
Factorized Transition Function: As Bayesian Network



Each variable evolution is dependant from few parents
Potentially with intermediate variables.

Factorized Transition Function: Dynamic Bayesian Network

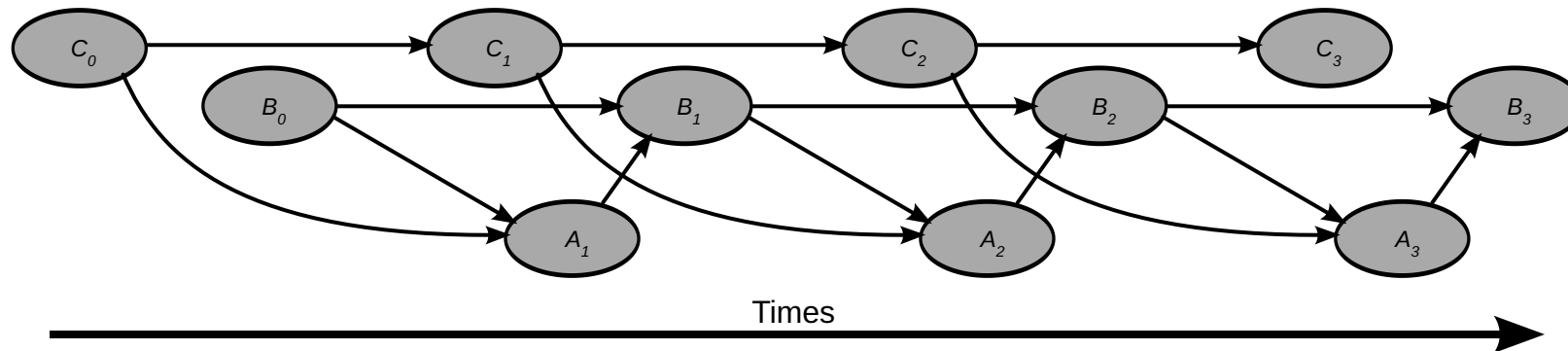
Bayesian Network with duplicated variables regarding their evolution in time.



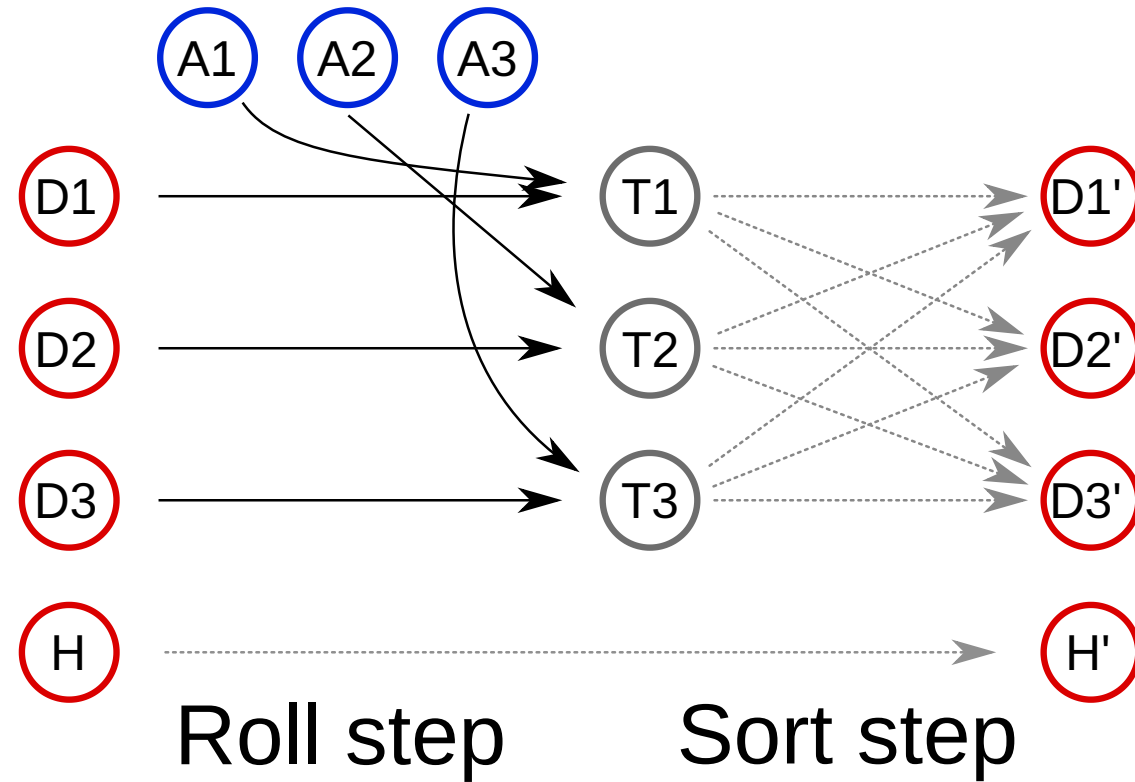
Considering a distribution of probability at time t ,
it is possible to infer over the distribution of probabilities at time $t + N$

Factorized Transition Function: Dynamic Bayesian Network

Extended definition over 3 times step.

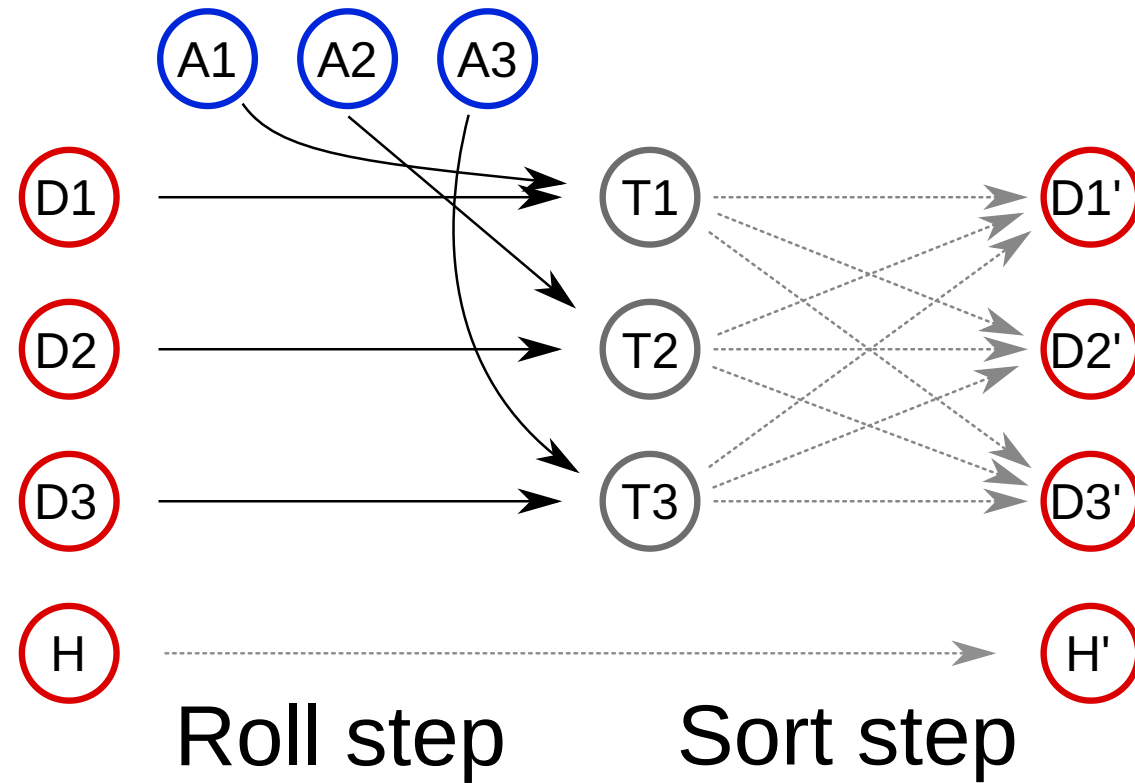


Factorized Transition Function: 421 example



- ▶ Roll step: 2 entrances - Small tables
- ▶ Sort step: 3 entrances but deterministic

Factorized Transition Function: 421 example



► Let consider cheat dice:

Learning transitions $\rightarrow$ computing 3×12 probabilities *vs* $(168^2 \times 8)$

Factorized Transition Function: Zombie Dice Example

Naturally the complexity of the networks grow linearly with the complexity of the game



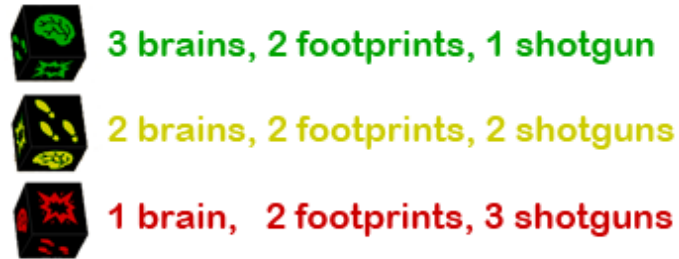
A stop and go game:

2 players or more (2 Zombies)

- The first one to cumulate 13 brains win.
- Player need to pass, before to get 3 damages

Factorized Transition Function: Zombie Dice Example

A 4 steps' game engine:



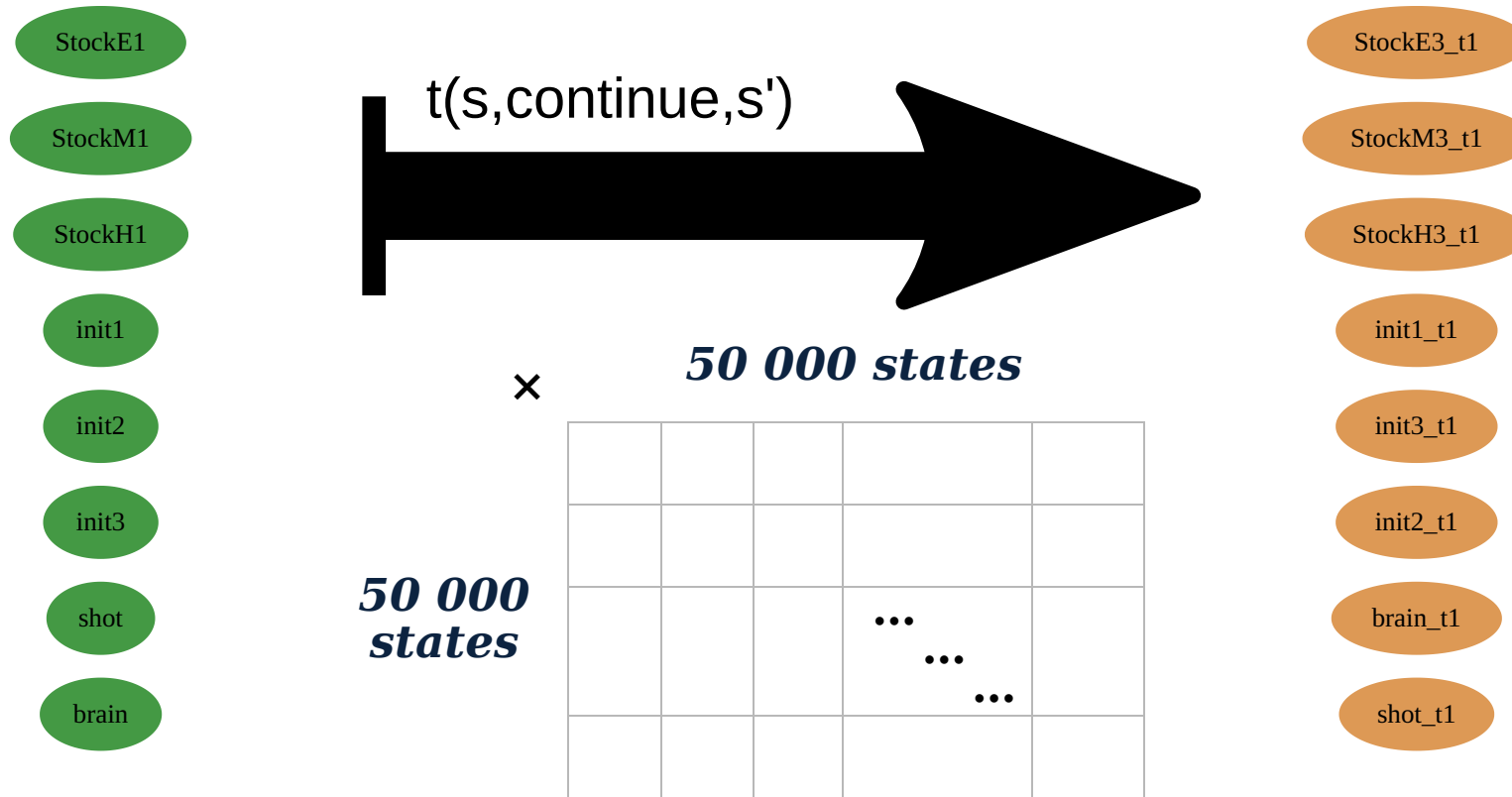
Complete Dice Set:



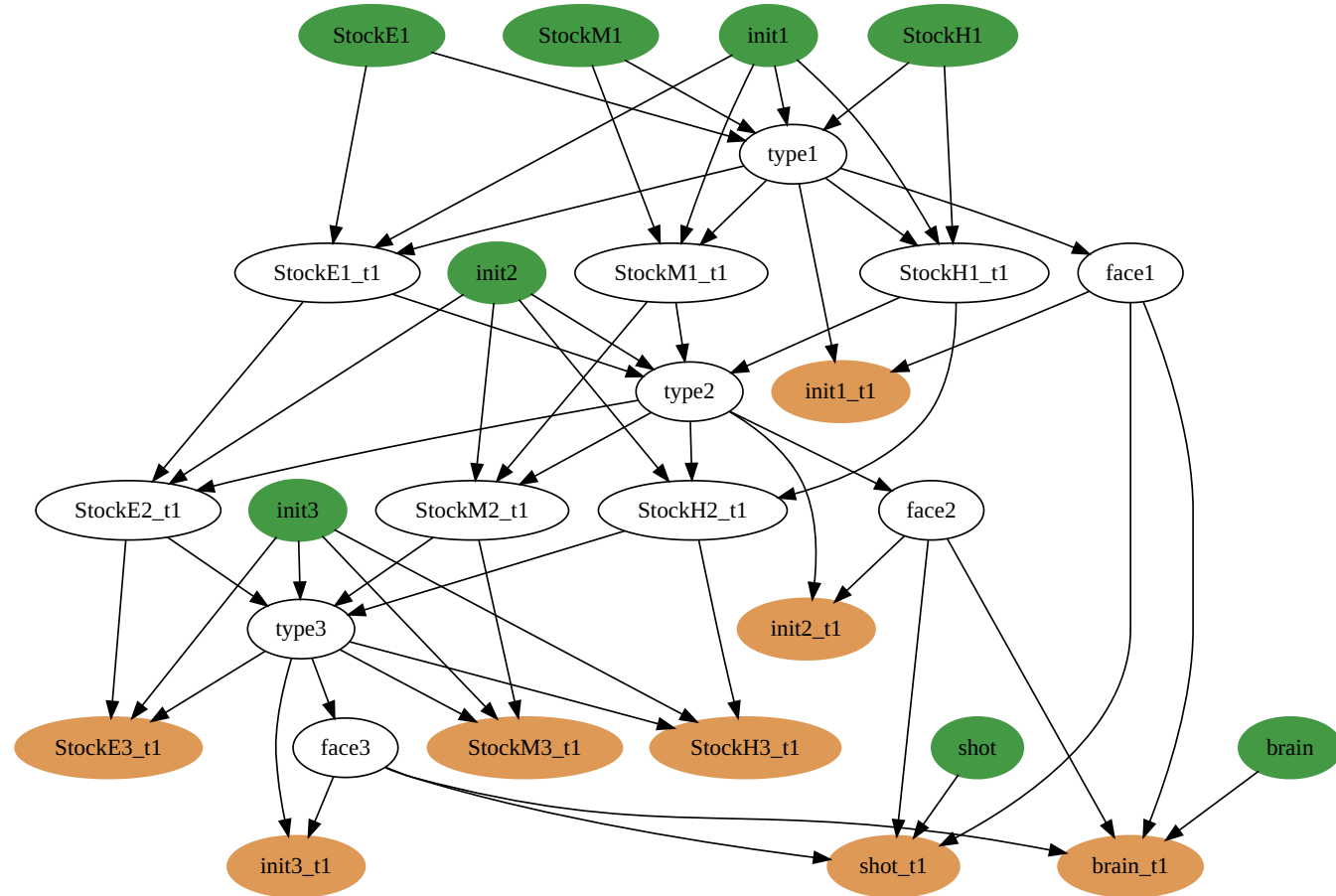
1. player decides to continue or not
2. new dice are picked randomly (to complete the hand)
3. dice are rolled
4. the player cumulates brains and damages

► State Space ? Branching ?

Zombie Dice : Brut Transition Function (action: go)



Zombie Dice : Factorized Transition Function (action: go)



Conclusion on Bayesian-network based transitions

- ▶ Bayesian Network allows for **efficient factorized transition** definition
- ▶ **Learning process:** (very hard)
 - Quid of network structure ? (correlation is not causality)
([recent survey](#))
 - And with hidden variables ?
- ▶ Going futher: [distribution models](#)...